

PART V

Optimizing Performance in Sport

Previous chapters explained how the body responds to an acute bout of exercise, how it adapts to chronic training, and how it adjusts to environmental extremes. We can now apply that knowledge to optimizing performance specific to sport. In [part V](#), we focus on how athletes can best prepare for competition from a physiological standpoint. In [chapter 14](#), Training for Sport, we discuss how to optimize athletes' training regimen and explore how too little or too much training can impair performance. In [chapter 15](#), Body Composition and Nutrition for Sport, we address the issues of assessing body composition, how body composition relates to sport performance, and sports that have weight standards. We then evaluate athletes' dietary needs and consider how changing the diet or using nutritional supplementation may improve performance. In [chapter 16](#), Ergogenic Aids in Sport, we discuss many of the pharmacological, hormonal, and physiological agents that have been proposed to improve performance. We examine the potential benefits, proven effects, and health risks associated with their use.





14

Training for Sport

In this chapter and in the web study guide

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AUDIO FOR FIGURE 14.6 describes a model for the development of overtraining and the roles of the brain and immune system.

Tapering for Peak Performance

VIDEO 14.1 presents Scott Trappe on tapering for peak sport performance.

ACTIVITY 14.2 *Tapering for Peak Performance* investigates the effects of tapering on the performance of two distance runners.

Detraining

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ACTIVITY 14.3 *Detraining* explores how physiological responses to detraining change.

In Closing

Throughout his college career, Eric had trained at swimming 4 h each day, covering as much as 13.7 km (8.5 mi) per day. Despite this effort, his performance time for the 200 yd (183 m) butterfly event had not improved since his freshman year. With a best performance of 2 min 15 s for the event, he was seldom given a chance to compete because several teammates could perform the event in less than 2 min 5 s. During Eric's senior year, his coach made a major change in the team's training plan. The swimmers trained only 2 h per day and swam an average of 4.5 to 4.8 km (2.8-3.0 mi) per day. Further, they swam each interval at a faster pace and had a longer rest period between intervals. Suddenly Eric's performance began to improve. After 3 months, his time dropped to 2 min 10 s, still not good enough to make him a major contender. But as a reward for Eric's improvement, the coach chose him to swim the 200 yd butterfly event at the conference championship meet, which was preceded by 3 weeks of reduced training (tapering) of only 1.6 km (1 mi) per day. Subsequently, with less volume of training than in previous years and well rested after the taper, Eric was able to make the finals of the event at the championship meet. His preliminary time was 2 min 1 s. In the finals, he improved even further, posting a third-place finish with a time of 1 min 57.7 s, an impressive performance for a swimmer who performed better with lower-volume but higher-quality training.

Repeated days and weeks of training can be considered a positive stress because the adaptations caused by training improve the capacity for energy production, oxygen delivery, muscle contraction, and other mechanisms that enhance exercise performance. The major changes associated with training occur within the first 6 to 10 weeks. The magnitude of these adaptations depends on the volume and intensity of exercise performed during training, which has led many coaches and athletes to believe erroneously that the athlete who trains the longest and hardest will be the best performer. However, the quantity and the quality of training are two separate things. Too often, training sessions are judged by the total volume (e.g., the distance run, cycled, or swum) performed in each training session, leading coaches to design training programs that are not optimal for improving performance and often impose unrealistic demands on the athlete.

The rate at which an individual adapts to training is genetically limited. Too much training can reduce the athlete's optimal potential for improvement and in some cases can cause a breakdown in the adaptation process, eventually reducing performance. When training is taken to extremes, serious illness or injury can occur.

Although the volume of work performed during training is an important stimulus for physiological improvements in performance, a proper balance should be established between volume and intensity. Training can be overdone, leading to fatigue, illness, overuse injury, overtraining syndrome, and performance decrements. In contrast, proper rest and achieving the proper balance between training volume and intensity can, and will, enhance performance. Much effort has been directed toward determining the appropriate volume and intensity required to achieve optimal adaptation. Exercise physiologists have tested numerous training regimens to determine both the minimal and maximal stimuli needed for cardiovascular and muscular improvements. The next section examines those factors that can affect the response to a given training program, developing a model for optimizing the training stimulus.

Optimizing Training

All well-designed training programs incorporate the principle of progressive overload. As discussed in [chapter 9](#), this principle states that to continue to provide the benefits of training, the training stimulus must be progressively increased as the body adapts to the current stimulus. The only way to continue to improve with training is to progressively increase the training stimulus. However, when this concept is carried too far, training may become excessive, pushing the body beyond its ability to adapt, producing no additional improvement in conditioning or performance and leading to performance decrements. Conversely, if the volume or the intensity of training is too low, the resulting physiological changes will be hindered and optimal performance will not be achieved. Thus, the coach and athlete face the challenge of determining the optimal training stimulus for each particular athlete, recognizing that what works for one athlete might not work for another.

Figure 14.1 provides a model demonstrating the continuum of training stages. In this model, **undertraining** represents the type of training an athlete would undertake between competitive seasons or during active rest. Generally, physiological adaptations will be minor, and there will be no improvement in performance during this stage. **Acute overload** represents what might be considered an average training load, whereby the athlete is stressing the body to the extent necessary to improve both physiological function and performance. **Overreaching** is a relatively new term that refers to a brief period of heavy overload without adequate recovery, thus exceeding the athlete's adaptive capacity. There will be a brief performance decrement, from several days to several weeks, but eventually performance will improve. Finally, **overtraining** refers to that point at which an athlete experiences physiological maladaptations and chronic performance decrements. This generally leads to the **overtraining syndrome**.¹

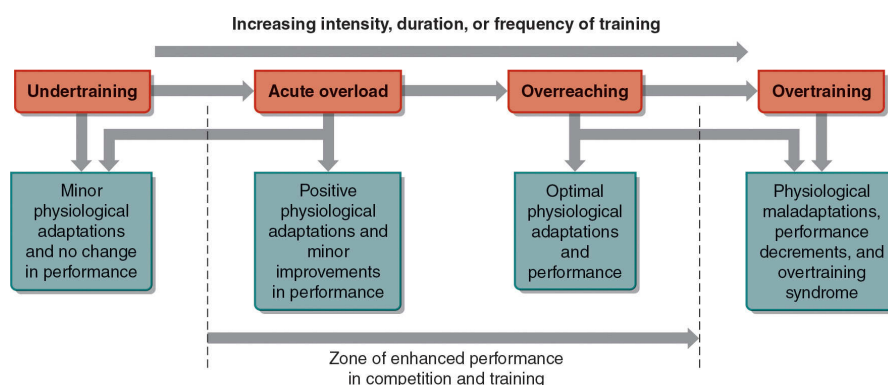


FIGURE 14.1 Model of the continuum of training stages.

Adapted by permission from L.E. Armstrong and J.L. VanHeest, "The Unknown Mechanism of the Overtraining Syndrome," *Sports Medicine* 32, no. 1 (2002): 185-209.

Overreaching

Overreaching is a systematic attempt to intentionally overstress the body for a short period of training. Done correctly, this allows the body to adapt to the increased training stimulus, that is, beyond the level of adaptation attained during normal overload. As with overtraining, there is a brief decrement in performance lasting several days to several weeks, followed by increased physiological function and increased performance. Obviously, this is the critical phase of training

—on the edge, leading either to improved physiological function and performance or to overtraining if one goes too far. With overreaching, the period of full recovery from training takes several days to several weeks; however, with overtraining, recovery can take many months or, in some cases, years. The key to overreaching is to push the athlete hard enough to accomplish the desired positive physiological and performance improvements but to avoid going into the stage of overtraining. This is not an easy task to accomplish!

Excessive Training

Although not shown in the model in [figure 14.1](#), **excessive training** refers to training that is well above what is needed for peak performance but does not strictly meet the criteria for either overreaching or overtraining. With excessive training, either the volume or the intensity of training, or both, are increased beyond the optimal level. A “more is better” philosophy often drives the training schedule. For many years, athletes were undertrained. As coaches and athletes became bolder and started to push the envelope by increasing both training volume and intensity, they found that athletes responded well, and world records began to tumble. However, one can take this philosophy only so far. At a certain point, performance begins to either plateau or decline.

Most research on excessive training has been conducted on swimmers, but the principles also apply to most other forms of training. Swim training 3 to 4 h per day, 5 or 6 days each week, provides no greater benefits than training only 1 to 1.5 h per day.⁶ In fact, such excessive training has been shown to significantly decrease muscular strength and sprint swimming performance.

Studies conducted to date reveal no scientific evidence that multiple daily training sessions enhance fitness and performance more than a single daily session. This is illustrated by the data in [figure 14.2](#), which show the responses of two groups of swimmers who trained once per day (group 1) or twice per day (group 2) for weeks 5 through 10 of a 25-week training program.⁶ All swimmers began the program following the same training regimen: one session per day. But from the beginning of the 5th week through the end of the 10th week, group 2 trained twice per day, doubling their volume of training. After 6 weeks on the different regimens, both groups

returned to the once-daily program. All the swimmers' heart rates and blood lactate values decreased dramatically when training began, and no significant differences were seen in the two groups' results in response to the change in training volume. The swimmers who trained twice per day showed no additional improvements over those who trained only once per day. In fact, their blood lactate concentrations ([figure 14.2a](#)) and heart rates ([figure 14.2b](#)) appeared to be slightly higher for the same fixed-pace swim.

To determine the influence of long-term excessive training, performance improvements of swimmers who trained twice daily for a total distance of more than 10,000 m (10,936 yd) per day (the LS, or long-swim, group) were compared with improvements of those who swam approximately half that distance in a single session each day (the SS, or short-swim, group).⁶ Changes in performance time for the 100 yd (91 m) front crawl were examined over a 4-year period for both groups. The LS swimmers and SS swimmers experienced an identical improvement of 0.8% per year. The concept of training specificity (see [chapter 9](#)) implies that several hours of daily training will not provide the adaptations needed for athletes who participate in events of short duration. Most competitive swimming events last less than 2 min. How can training for 3 to 4 h per day at speeds that are markedly slower than competitive pace prepare the swimmer for the maximal efforts of competition? Such a large training volume prepares the athlete to tolerate a high volume of training but likely does little to benefit actual performance.

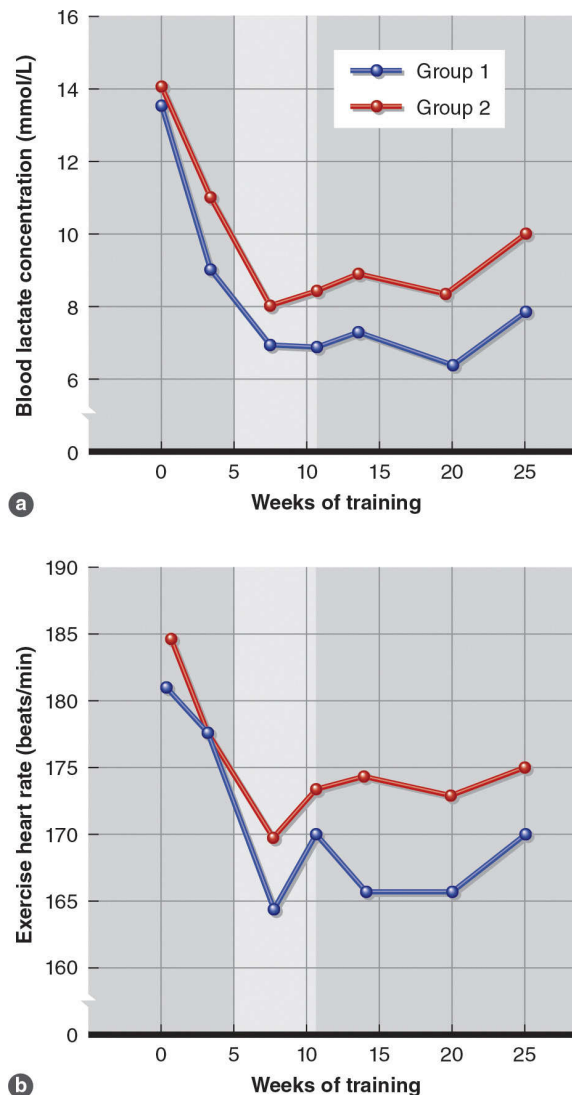


FIGURE 14.2 Changes in swimmers' (a) blood lactate concentrations and (b) heart rates during a standardized 366 m (400 yd) swim over 25 weeks of training. From the beginning of the fifth through the end of the 10th week, one group trained once per day (group 1) whereas the other group trained twice each day (group 2).

The need for long daily workouts (high *volume*) is now being questioned by researchers. For many sports, it appears that training volume could be reduced significantly, possibly by as much as one-half in some sports, without reducing the benefits and with less risk of overtraining athletes to the point of decreased performance. The principle of training specificity suggests that low-intensity, high-volume training does not improve sprint-type performance.

Training *intensity* is also an important factor and refers to both the relative force of muscle action (i.e., resistance training) and the

relative stress placed on the metabolic and cardiovascular systems (i.e., anaerobic and aerobic training). There is a strong interaction between training intensity and training volume: As intensity is reduced, training volume must be increased to achieve adaptation. Training at very high intensities requires substantially less training volume, but the adaptations that occur will be significantly different from those achieved with low-intensity, high-volume training. This concept applies to all three types of training, that is, resistance, anaerobic, and aerobic.

High-intensity, low-volume training can be tolerated only for brief periods. While this type of training does increase muscular strength and anaerobic capacity in high-intensity interval training, it provides little or no improvement in aerobic capacity. Conversely, low-intensity, high-volume training stresses the oxygen transport and oxidative metabolic systems, causing greater gains in aerobic capacity, but has little or no effect on muscular strength, anaerobic capacity, or sprint speed.

Attempts to perform large volumes of high-intensity training can have negative effects on adaptation. The energy needs of high-intensity exercise place greater demands on the glycolytic system, rapidly depleting muscle glycogen. If such training is attempted too often—for example, daily—the muscles can become chronically depleted of their energy reserves, and the person might demonstrate signs of overtraining, a topic discussed later in this chapter.

Periodization of Training

Based on the model of training shown in [figure 14.1](#), effective training is accomplished by applying the principle of **periodization**, dividing the entire sport training season into smaller periods of time and training units. Periodization allows for a varied training load over time that enables acute overload and overreaching without overtraining the athlete. Traditional periodization began in the 1960s and gained popularity among coaches and athletes over the next five decades. More recently, alternative approaches have appeared, but little research has been performed to validate either the physiological adaptations or sport performance improvements that accompany them.

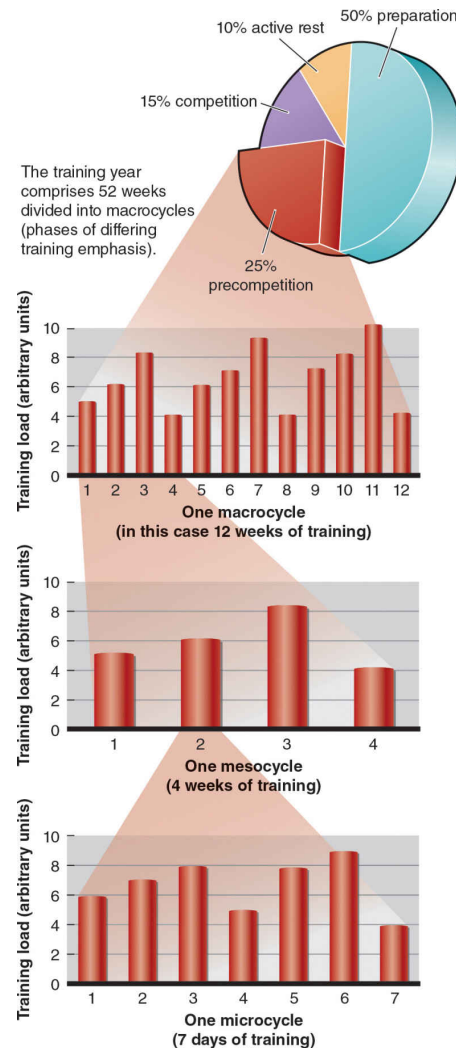
Traditional Periodization

Periodization, in its traditional form, involved the planned sequencing of training units (longer and shorter sessions and cycles, higher and lower intensity periods) so that athletes could achieve their desired performance. For decades, periodization was based on a hierarchy that included

- multiyear preparation (for example, for the 4-year Olympic cycle),
- macrocycles (monthlong training cycles),
- mesocycles (weekly training cycles),
- microcycles (cycle lasting a few days), and finally,
- individual workouts.

One such periodized training program is illustrated in [figure 14.3](#).

Traditional periodization programs have two major disadvantages.¹⁴ First, models such as that shown in [figure 14.3](#) are best used to prepare athletes to participate successfully in a single planned competition. The recent trend for high-caliber athletes to compete in multiple competitions throughout the year adds too much complexity to the traditional design of macro-, meso-, and microcycles. Secondly, the traditional periodization scheme assumes that all physiological systems and skills involved in the sport are being developed simultaneously. This approach works best for sports like running, swimming, and cycling but falls short for ball sports, combat sports, and aesthetic sports that require more than the development of general attributes like aerobic capacity, muscular strength, or sprint speed.



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FIGURE 14.3 The structure of a traditional periodized training program. This model varies the training load over time to achieve acute overload and some overreaching while avoiding overtraining.

Adapted by permission from R.W. Fry, A.R. Morton, and D. Keast, "Overtraining in Athletes: An Update," *Sports Medicine* 12 (1991): 32-65.

An effective variation of this model involves the implementation of general and specific exercises with the goal of stimulating specific motor skills that will stress both metabolic and sport-specific movement patterns. General exercises aim at developing basic motor abilities (e.g., general strength, speed, or endurance) regardless of the mechanical or metabolic demands of the sports, for example, squats for soccer. Specific exercises are designed to mimic mechanical gestures (movement patterns) specific to a sport, for example, lunges for fencing. Specificity can be also achieved by

stressing the *most relevant* physiological systems for a given sport, with special emphasis on intensity and duration. By combining the neuromuscular and mechanical with the metabolic components, sport-specific exercise can be optimized for almost every sport.

RESEARCH PERSPECTIVE 14.1

Periodization of High-Intensity Training Models and Endurance Adaptation

The training term *periodization* is widely used to describe and quantify the planning process of training. Periodization plans add structure, with well-defined training periods that stimulate specific physiological adaptations or performance qualities for optimal performance development. Such endurance training models manipulate different training sessions periodized over time, scales ranging from microcycles (2-7 days), mesocycles (3-6 weeks), and macrocycles (6-12 months). Training intensity distribution is also considered, with most studies suggesting a high volume of low-intensity training, substantial moderate-intensity training, and less high-intensity training. The optimization of the duration and intensity of individual training sessions, the frequency of training sessions, and the organizational pattern of these stimulus variables will maximize physiological adaptations and performance capability in elite athletes.

Research has recently focused on the periodization of high-intensity training. Although these limited studies have advanced our understanding of the intensity and duration of high-intensity training sessions and endurance performance, the cumulative effects of the intensity and duration of such training sessions is not completely understood. Thus, investigators undertook a study to compare the effects of three different high-intensity training models, balanced for total load but periodized in a specific mesocycle order or in a mixed distribution, on endurance adaptations during a 12-week training period in endurance athletes.²¹ Three separate groups were assigned to organized interval sessions in a specific periodized mesocycle order (increasing or decreasing intensity) or in a mixed distribution during mesocycles. This training intervention is illustrated in the accompanying figure. Pre- and postintervention testing assessed both physiological (e.g., $\dot{V}O_2$) and performance (e.g., power) responses.

The results showed that all three groups demonstrated moderate-to-large improvements in physiological and performance variables, indicating that the basic load features of all three training models were effective. Perhaps surprisingly, the specific high-intensity training periodized mesocycle order or mixed distribution did not have any effect on the adaptive responses to endurance training. Although the authors found a tendency toward less adaptation in the mixed distribution group, this was not statistically different.

The authors argue that many different soundly designed periodization structures can achieve the same training goals.

A key to determining the effectiveness of periodization programs is to test performance improvements following each cycle using sport-specific drills. Tests need to be specific to the sport for which training is implemented so that a true performance improvement can be assessed. Implementing this approach will serve as the basis for the coming training cycles. With appropriate testing following each training cycle, it is possible to determine not only performance improvements but also any transfer effects from the general to the specific training modules or cycles.

Block Periodization

Beginning in the 1980s, the term *training blocks* became popular among highly trained athletes and their coaches.¹⁴ A block is a training cycle of highly specialized and concentrated work. While each sport differs, block periodized training involves several general principles.¹⁴

- Each block focuses on a minimal number of desired outcomes and targeted abilities.
- The number of blocks is small (three or four), unlike traditional mesocycles that may contain 9 to 11 different types of smaller cycles.
- A single block typically lasts 2 to 4 weeks. This allows for adaptation without excessive fatigue.
- Putting training stages into the proper sequence leads to optimal competitive sport performance.

The major purported benefit of block periodization is the consecutive development of many abilities related to the given sport, although the published data on this topic are scarce because it is relatively new and results are difficult to compare across sports and studies.

In Review

- Every athlete's rate of adaptation to training is genetically determined. Each individual responds differently to the same training stress, so what might be

excessive training for one person could be well below the capacity of another for optimal adaptations. Therefore, it is important to recognize individual differences when designing training programs.

- Optimal training involves following a model that incorporates the principles of periodization, because the body needs to systematically go through stages of undertraining, acute overload, and overreaching to maximize performance.
 - Excessive training refers to training that is done with an unnecessarily high volume, intensity, or both. It provides few or no additional improvements in conditioning or performance and can lead to decreased performance and health problems.
 - Training volume can be increased through increase in the duration or frequency of training bouts or both. Many studies have shown no significant differences in improvement between athletes who train with typical training volumes and those who train with twice the volume (training conducted for twice the duration or twice a day instead of once a day).
 - Training intensity determines the specific adaptations that occur in response to the training stimulus. As training intensity increases, training volume must be reduced, and vice versa.
 - Periodization allows for a varied training load over time that allows for acute overload and overreaching without overtraining the athlete.
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Overtraining

With overly intense training, athletes may experience an unexplained decline in performance and physiological function that extends over weeks, months, or years. This condition is termed overtraining, and the precise cause or causes of the resulting performance decrement are not fully understood. Research has pointed to both psychological and physiological causes. Furthermore, overtraining can occur with each of three major forms of training—resistance, anaerobic, and aerobic—so it is likely that the cause or causes and symptoms vary by the type of training.

All athletes experience some degree of fatigue during repeated days and weeks of training, so not all fatigue-producing situations can be classified as overtraining (as noted previously with overreaching). Fatigue that follows one or more exhaustive training sessions is typically relieved by a few days of reduced training or rest and a

proper calorie- and carbohydrate-rich diet. Overtraining, on the other hand, is characterized by a sudden decline in performance and physiological function that cannot be remedied by a few days of reduced training, rest, or dietary intervention.

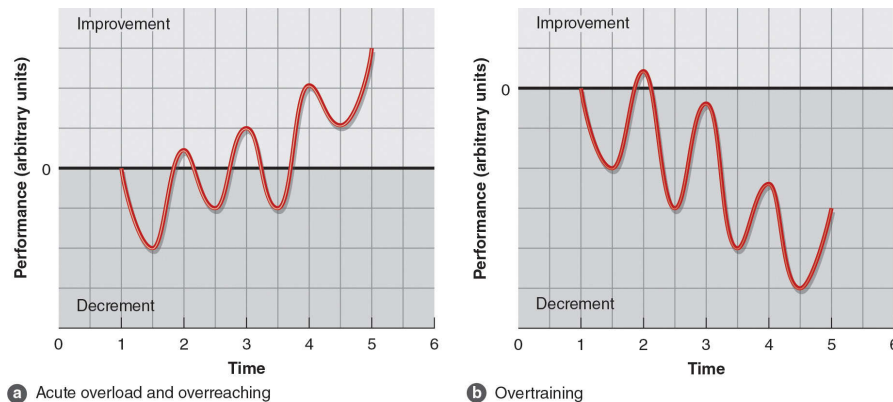


FIGURE 14.4 Typical pattern of the expected improvement in performance with (a) acute overload and overreaching in contrast to the pattern seen with (b) overtraining.

Adapted by permission from M.L. O'Toole, Overreaching and Overtraining in Endurance Athletes, in *Overtraining in Sport*, edited by R.B. Kreider, A.C. Fry, and M.L. O'Toole (Champaign, IL: Human Kinetics, 1998), 10, 13.

The Overtraining Syndrome

Most of the symptoms that result from overtraining, collectively referred to as the overtraining syndrome, are subjective and identifiable only after the individual's performance and physiological function have suffered. Unfortunately, these symptoms can be highly individualized, making it very difficult for athletes, trainers, and coaches to recognize that performance decrements were indeed caused by overtraining. A decline in physical performance with continued training is usually the first indication of the overtraining syndrome (see [figure 14.4](#)). The athlete senses a loss of muscular strength, coordination, and exercise capacity and generally feels fatigued. Other primary signs and symptoms of the overtraining syndrome include¹

- change in appetite;
- body weight loss;
- sleep disturbances;
- irritability, restlessness, excitability, and anxiousness;
- loss of motivation and vigor;

- lack of mental concentration;
- feelings of depression; and
- lack of appreciation for things—including exercise—that normally are enjoyable.

The underlying causes of overtraining syndrome are often a complex combination of emotional and physiological factors. The emotional demands of competition, the desire to win, the fear of failure, unrealistically high goals, and others' expectations can be sources of intolerable emotional stress. Because of this, overtraining is typically accompanied by a loss of competitive desire and a loss of enthusiasm for training. Furthermore, Armstrong and VanHeest¹ make the important observation that the overtraining syndrome and clinical depression involve remarkably similar signs and symptoms, brain structures, neurotransmitters, endocrine pathways, and immune responses, suggesting that they may have similar etiologies.

The physiological factors responsible for the detrimental effects of overtraining are likewise not fully understood. However, research suggests that overtraining is associated with alterations in the nervous, endocrine, and immune systems. Although a cause-and-effect relationship between these changes and the symptoms of overtraining has not been clearly established, these symptoms can help determine whether an individual is overtrained. In the following discussion, we focus on some of the observed changes associated with overtraining and on potential causes of the overtraining syndrome.

Autonomic Nervous System Responses to Overtraining

Some studies suggest that overtraining is associated with abnormal responses of the autonomic nervous system. Physiological symptoms accompanying the decline in performance often reflect changes in those organs or systems that are controlled by either the sympathetic or the parasympathetic branch of the autonomic nervous system (see [chapter 3](#)). Sympathetic nervous system abnormalities due to overtraining can lead to

- increased resting heart rate,
- increased blood pressure,

- loss of appetite,
- decreased body mass,
- sleep disturbances,
- emotional instability, and
- elevated basal metabolic rate.

This form of overtraining occurs predominantly among athletes who emphasize resistance training methods of very high intensity.

Other studies suggest that the parasympathetic nervous system might be dominant in some cases of overtraining, usually in endurance athletes. In these cases, the performance decrements markedly differ from those associated with sympathetic overtraining. Signs of parasympathetic overtraining, assumed to be the result of high training volume overload, include

- early onset of fatigue,
- decreased resting heart rate,
- rapid heart rate recovery after exercise, and
- decreased resting blood pressure.

Thus, it appears that athletes in different sports or events will likely exhibit unique signs and symptoms of the overtraining syndrome that are related to their training regimens. In fact, some have differentiated intensity-related and volume-related forms of overtraining, recognizing that specific training stressors result in unique signs and symptoms when applied excessively.

The symptoms of overtraining syndrome are highly individualized and subjective, so they cannot be universally applied. The presence of one or more of these symptoms is sufficient to alert the coach or trainer that an athlete might be overtraining.

Some of the symptoms associated with autonomic nervous system overtraining are also seen in people who are not overtrained. For this reason, the presence of these symptoms cannot be used to confirm overtraining. Although there is not strong scientific evidence to support the autonomic nervous system overtraining theory, the autonomic nervous system is definitely affected by overtraining.

Hormonal Responses to Overtraining

Measurements of various blood hormone concentrations during periods of overreaching suggest that marked disturbances in endocrine function accompany excessive stress. As shown in [figure 14.5](#), when swimmers increase their training 1.5- to 2-fold, blood concentrations of thyroxine and testosterone usually decrease and blood concentrations of cortisol increase. The ratio of testosterone to cortisol is thought to regulate anabolic processes in recovery, so a change in this ratio is considered an important indicator, and perhaps a cause, of the overtraining syndrome. Decreased testosterone coupled with increased cortisol might lead to more protein catabolism than anabolism in the cells. Other research, however, suggests that although cortisol concentrations increase with overreaching and the early stages of overtraining, both resting and exercise cortisol concentrations eventually decrease in the overtraining syndrome. Further, most overtraining studies have been conducted on aerobically trained endurance athletes. Fewer studies exist on anaerobically trained and resistance-trained athletes. Using the terminology introduced in the preceding section, intensity-related overtraining (anaerobic and resistance training) does not appear to alter resting hormonal concentrations.

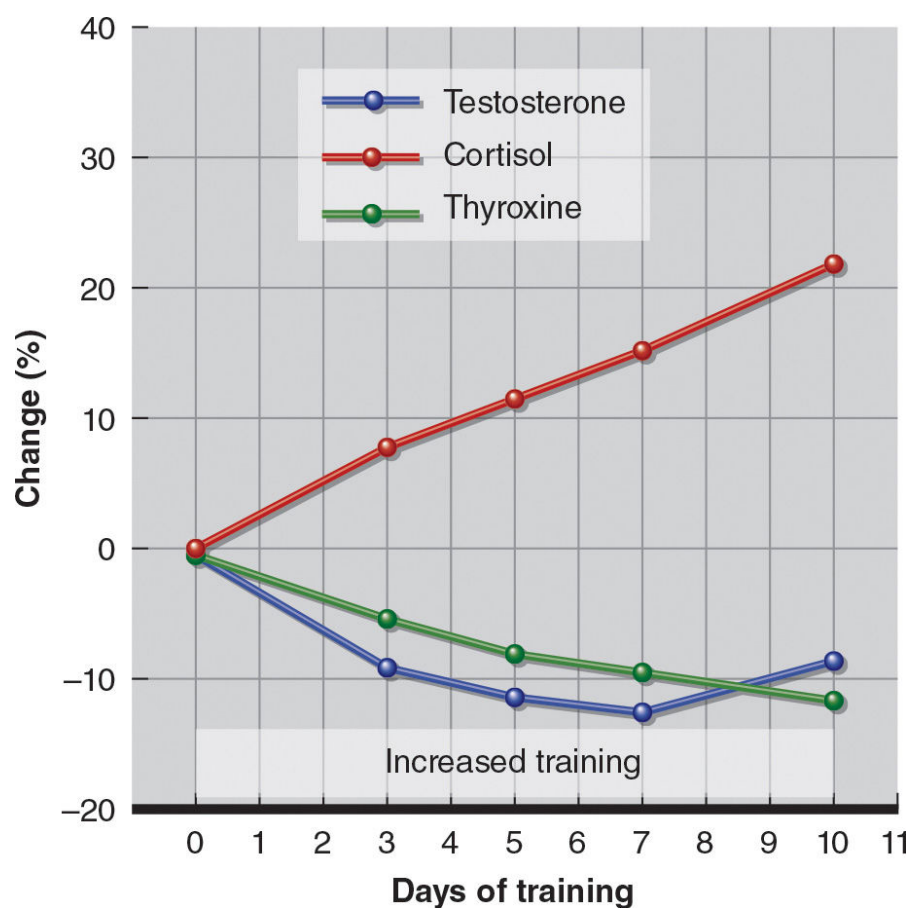


FIGURE 14.5 Changes in resting blood concentrations of testosterone, cortisol, and thyroxine over a period of intensified training. During the 10-day period shown here, swimmers increased their training from about 4,000 m (4,374 yd) to 8,000 m (8,749 yd) per day. These data show that resting cortisol concentrations increased in response to the added stress, whereas testosterone and thyroxine showed an unwanted decline during this period.

Overtrained athletes often have higher blood concentrations of urea because urea is produced by the breakdown of protein, that is, protein catabolism. This protein loss is the mechanism that is thought to be responsible for the weight loss seen in many overtrained athletes.

Elevated resting blood concentrations of epinephrine and norepinephrine have also been reported during periods of intensified aerobic training. These two hormones elevate heart rate and blood pressure. This has led some exercise physiologists to suggest that the blood concentrations of these catecholamines could be measured to confirm overtraining. However, other studies have found no change in these catecholamines during intensified training, and some have even found decreased resting concentrations.

Acute overload training and overreaching can produce most of the same hormonal changes reported in overtrained athletes. For this reason, measuring these and other hormones does not provide valid confirmation of overtraining. Athletes whose hormone concentrations appear to be abnormal may simply be experiencing the normal effects of training. Further, the time interval between the last training bout and the resting blood sample is very important. Some potential markers remain elevated for more than 24 h and might not reflect a true resting state. These hormonal changes might simply reflect the stress of training rather than a breakdown in the adaptive process. Consequently, experts generally agree that no blood marker conclusively defines the overtraining syndrome.

Armstrong and VanHeest¹ proposed that the various stressors associated with the overtraining syndrome act primarily through the hypothalamus. They postulated that these stressors activate the following two predominant hormonal axes involved in the body's response to stressors:

- The sympathetic–adrenal medullary axis (SAM), involving the sympathetic branch of the autonomic nervous system
- The hypothalamic-pituitary-adrenocortical axis (HPA)

This is illustrated in [figure 14.6a](#). [Figure 14.6b](#) illustrates the brain and immune system interactions with these two axes. These two figures are quite complex and go well beyond the scope of an introductory-level exercise physiology text. However, a cursory study of the interactions depicted in these figures will give one an appreciation of the complexity of this syndrome. Importantly, note that the stressors have their initial effect on the brain (hypothalamus). Thus, it is highly likely that brain neurotransmitters play an important role in the overtraining syndrome. Serotonin is a major neurotransmitter that is suspected to play a significant role in the overtraining syndrome. Unfortunately, plasma concentrations of this important neurotransmitter do not accurately reflect concentrations in the brain. Advances in technology should provide the necessary tools to help us better understand what is going on inside the brain.

Overtraining syndrome appears to be associated with systemic inflammation and increased synthesis of cytokines,²⁰ providing

support for the model depicted in [figure 14.6b](#). Elevated circulating cytokines result from infection as well as from skeletal muscle, bone, and joint trauma associated with overtraining. They appear to be a normal part of the body's inflammatory response to infection and injury. It is theorized that excessive musculoskeletal stress, coupled with insufficient rest and recovery, sets up a cascade of events whereby a local acute inflammatory response evolves into chronic inflammation and eventually into systemic inflammation. Systemic inflammation activates circulating monocytes, which can then synthesize large quantities of cytokines. Cytokines then act on most of the brain and body functions in a manner consistent with symptoms expressed in the overtraining syndrome.²⁰

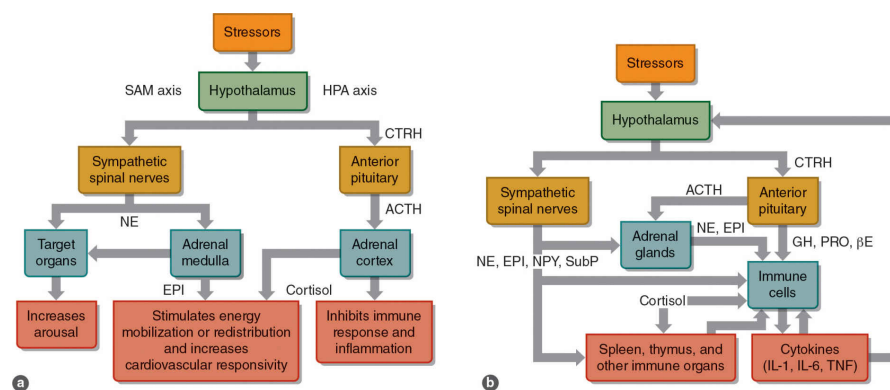


FIGURE 14.6 (a) The role of the hypothalamus and the sympathetic–adrenal medullary (SAM) and hypothalamic–pituitary–adrenocortical (HPA) axes as possible mediators of the overtraining syndrome. (b) The brain–immune system interactions in this model, with the cytokines playing a potentially major role in mediating overtraining. ACTH = adrenocorticotropin; EPI = adrenaline (epinephrine); CTRH = corticotrophin-releasing hormone; GH = growth hormone; IL-1 = interleukin-1; IL-6 = interleukin-6; NE = noradrenaline; SubP = substance P; PRO = prolactin; TNF = tumor necrosis factor; NPY = neuropeptide Y; β E = β -endorphin.

Adapted by permission from L.E. Armstrong and J.L. VanHeest, “The Unknown Mechanism of the Overtraining Syndrome,” *Sports Medicine* 32 (2002): 185-209.



Immunity and Overtraining

The immune system provides a line of defense against invading bacteria, parasites, viruses, and tumor cells. This system depends on the actions of specialized cells (such as lymphocytes, granulocytes, and macrophages) and antibodies. These primarily eliminate or neutralize foreign invaders that might cause illness (pathogens). Unfortunately, one of the most serious consequences of overtraining is the negative effect it has on the body's immune system. In fact, according to the model proposed in [figure 14.6](#), compromised **immune function** is potentially a major factor in the initiation of the overtraining syndrome.

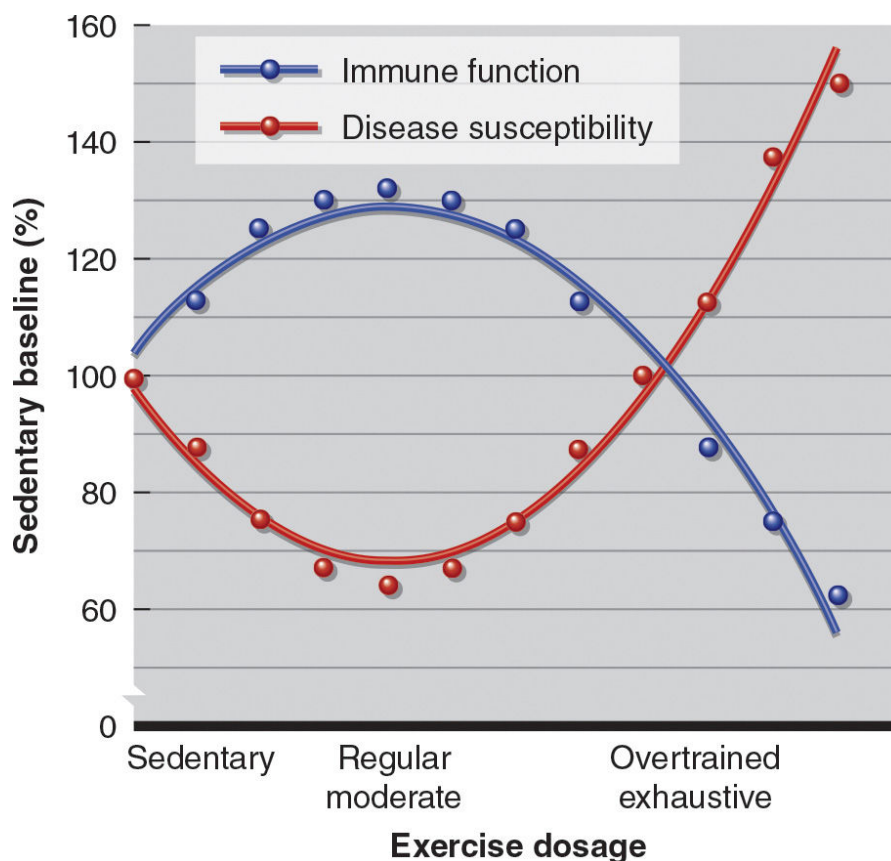


FIGURE 14.7 The inverted J-shaped model of the relationship between amount of exercise and immune function. This model suggests that moderate exercise may lower the risk of infection or disease, whereas overtraining may increase the risk.

Data from Nieman (1997).

Many studies have shown that excessive training suppresses normal immune function, increasing the overtrained athlete's susceptibility to infections. This is illustrated in [figure 14.7](#). Studies

also show that short bouts of intense exercise can temporarily impair the immune response, and successive days of heavy training can amplify this suppression. Several investigators have reported an increased incidence of illness following a single, exhaustive exercise bout, such as running a full 42 km (26.2 mi) marathon. This immune suppression is characterized by abnormally low concentrations of both lymphocytes and antibodies. Invading organisms or substances are more likely to cause illness when these concentrations are low. Also, intense exercise during illness might decrease one's ability to fight off the infection and increase the risk of even greater complications.¹⁸

Joint Consensus Statement on Overtraining

The importance of preventing, diagnosing, and treating the overtraining syndrome was addressed by two international organizations (the American College of Sports Medicine and the European College of Sport Science) in the form of a consensus statement.¹⁷ The balance between providing the overload needed for training adaptations and avoiding the combination of excessive overload and inadequate recovery is a difficult one to achieve. The consensus statement delineates between nonfunctional overreaching (NFOR) and overtraining syndrome (OTS). Nonfunctional overreaching leads to a state of stagnated or decreased performance, but eventually the athlete can recover with sufficient rest. Overtraining syndrome, on the other hand, results in a “long-term decrement in performance capacity with or without related physiological and psychological signs and symptoms of maladaptation in which restoration of performance capacity may take *several weeks or months*” [emphasis added] (p. 187). Survey data suggest that the prevalence of NFOR or OTS is approximately 10% in endurance athletes, but some estimates have been even higher.

The key distinction is that OTS results in prolonged maladaptation, but in reality it is extremely difficult to diagnose OTS. Symptoms of OTS are usually (but not always) more severe than those of NFOR and include fatigue, performance decline, and mood disturbances. Several markers of OTS have been proposed, including hormones, psychological tests, biochemical markers, and indices of immune function, but none of them meet the criteria to make their use

generally accepted. Measuring heart rate variability as an index of autonomic balance has shown some promise, but varied measurement techniques have limited its utility and acceptance. One necessary but tedious task in diagnosing OTS is to rule out other causes of performance decline such as diseases and infections, negative energy balance, insufficient carbohydrate or protein intakes, allergies, and so on.

Finally, it is important to consider that demanding, intensified training is not the only element in the development of OTS. Overtraining syndrome is complex and involves psychological factors such as excessive expectations, competitive stress, social and family involvement, personal and emotional problems, training monotony, and extra-sport demand from work or school. The two organizations propose the following checklist to aid in the diagnosis of OTS.

Performance-Fatigue

Is the athlete experiencing the following:

- Unexplainable underperformance
- Persistent fatigue
- Increased sense of effort in training
- Sleep disorders

Exclusion Criteria

Are there confounding diseases?

- Anemia
- Epstein-Barr virus
- Other infectious diseases
- Muscle damage (high CK)
- Lyme disease
- Endocrine diseases (diabetes, thyroid, adrenal gland, ...)
- Major disorders of eating behavior
- Biological abnormalities (increased erythrocyte sedimentation rate, C-reactive protein, creatinine, or liver enzymes, decreased ferritin, ...)
- Injury (musculoskeletal system)
- Cardiovascular symptoms

- Adult-onset asthma
- Allergies

Are there training errors?

- Training volume increased ($>5\%$) ($\text{h} \cdot \text{wk}^{-1}$, $\text{km} \cdot \text{wk}^{-1}$)
- Training intensity increased significantly
- Training monotony present
- High number of competitions
- In endurance athletes: decreased performance at “anaerobic” threshold
- Exposure to environmental stressors (altitude, heat, cold, ...)

Other confounding factors:

- Psychological signs and symptoms (disturbed POMS, RESTQ-Sport, RPE, ...)
- Social factors (family relationships, financial, work, coach, team, ...)
- Recent or multiple time zone travel

Exercise test:

- Are there baseline values to compare with (performance, heart rate, hormonal, lactate, ...)?
- Maximal exercise test performance
- Submaximal or sport-specific test performance
- Multiple performance tests

Reprinted by permission from R. Meeusen et al., “Prevention, Diagnosis, and Treatment of the Overtraining Syndrome: Joint Consensus Statement of the European College of Sport Science and the American College of Sports Medicine,” *Medicine and Science in Sports and Exercise* 45 (2013): 186-205.

TABLE 14.1 Potential Markers of Overreaching (OR), Overtraining (OT), and Overtraining Syndrome (OTS)

Physiological and psychological marker	Response	OR	OT	OTS
HR _{rest} and HR _{max}	Decreased		✓	✓
HR _{submax} and $\dot{V}O_{2submax}$	Increased	✓		✓
$\dot{V}O_{2max}$	Decreased			✓
Anaerobic metabolism	Impaired		✓	
Basal metabolic rate	Increased			✓
RER _{submax} and RER _{max}	Decreased		✓	✓
Nitrogen balance	Negative			✓
Nerve excitability	Increased			✓
Sympathetic nervous response	Increased			✓
Psychological mood states	Altered	✓		
Risk of infection	Increased	✓		
Hematocrit and hemoglobin	Decreased		✓	
Leukocytes and immunophenotypes	Decreased		✓	
Serum iron and ferritin	Decreased		✓	
Serum electrolyte levels	Decreased			✓
Serum glucose and free fatty acids	Decreased		✓	
Plasma lactate concentration, submax, max	Decreased		✓	✓
Ammonia	Increased		✓	✓
Serum testosterone and cortisol	Decreased	✓		
ACTH, growth hormone, prolactin	Decreased			✓
Catecholamines, rest, night	Decreased			✓
Creatine kinase	Increased			✓

Note. HR = heart rate; RER = respiratory exchange ratio; ACTH = adrenocorticotrophic hormone.

From Armstrong and VanHeest (2002).

Predicting the Overtraining Syndrome

We must remember that the underlying cause or causes of the overtraining syndrome are not fully known, although it is likely that physical or psychological (or emotional) overload, or a combination of the two, might trigger this condition. Trying not to exceed an athlete's stress tolerance by regulating the amount of physiological and psychological stress experienced during training is difficult. Most coaches and athletes use intuition to determine training volume and intensity, but few can accurately assess the true impact of a workout on the athlete. No preliminary symptoms warn athletes that they are on the verge of becoming overtrained. By the time coaches realize that they have pushed an athlete too hard, it is often too late. The damage done by repeated days of excessive training or stress can be

repaired only by days, and in some cases weeks or months, of reduced training or complete rest.

Numerous investigators have tried to identify markers of the overtraining syndrome in its early stages by using assorted physiological and psychological measurements. A list of potential markers is provided in [table 14.1](#). Unfortunately, none has proven totally effective. It is often difficult to determine whether the measurements obtained are related to overtraining or whether they simply reflect normal responses to overload or overreaching training.

One good method to identify the overtraining syndrome is to monitor the athlete's heart rate during a standardized workout, such as a fixed-paced run or swim, using a digital heart rate monitor ([figure 9.5](#)). The data presented in [figure 14.8](#) illustrate a runner's heart rate response during a 1 mi (1.6 km) run performed at a fixed pace of 6 min/mi (3.7 min/km), or 10 mph (16 km/h). This response was monitored when the runner was untrained (UT), after the runner had trained (TR), and during a period when the runner demonstrated symptoms of overtraining syndrome (OT). This figure shows that heart rate was higher when the runner was in the overtrained state than when the runner was responding well to training. Such a test provides a simple and objective way to monitor training and can possibly provide an early warning sign of overtraining.

Exercise Addiction

The condition of exercise addiction may not sound like a bad thing to many people who understand all of the health benefits associated with regular exercise. However, exercise addiction is maladaptive, so instead of improving the person's life, it causes more problems. This form of addictive behavior can threaten one's overall health, cause overuse and other injuries, ruin personal lives and careers, and, in some cases, contribute to nutrition problems and eating disorders. Not surprisingly, exercise addiction has received recent attention among exercise scientists, including exercise and sport psychologists. Surprisingly, despite the known harmful consequences, exercise addiction is not listed as a disorder in the latest version of the *Diagnostic and Statistical Manual of Mental Disorders (DSM-V)*.

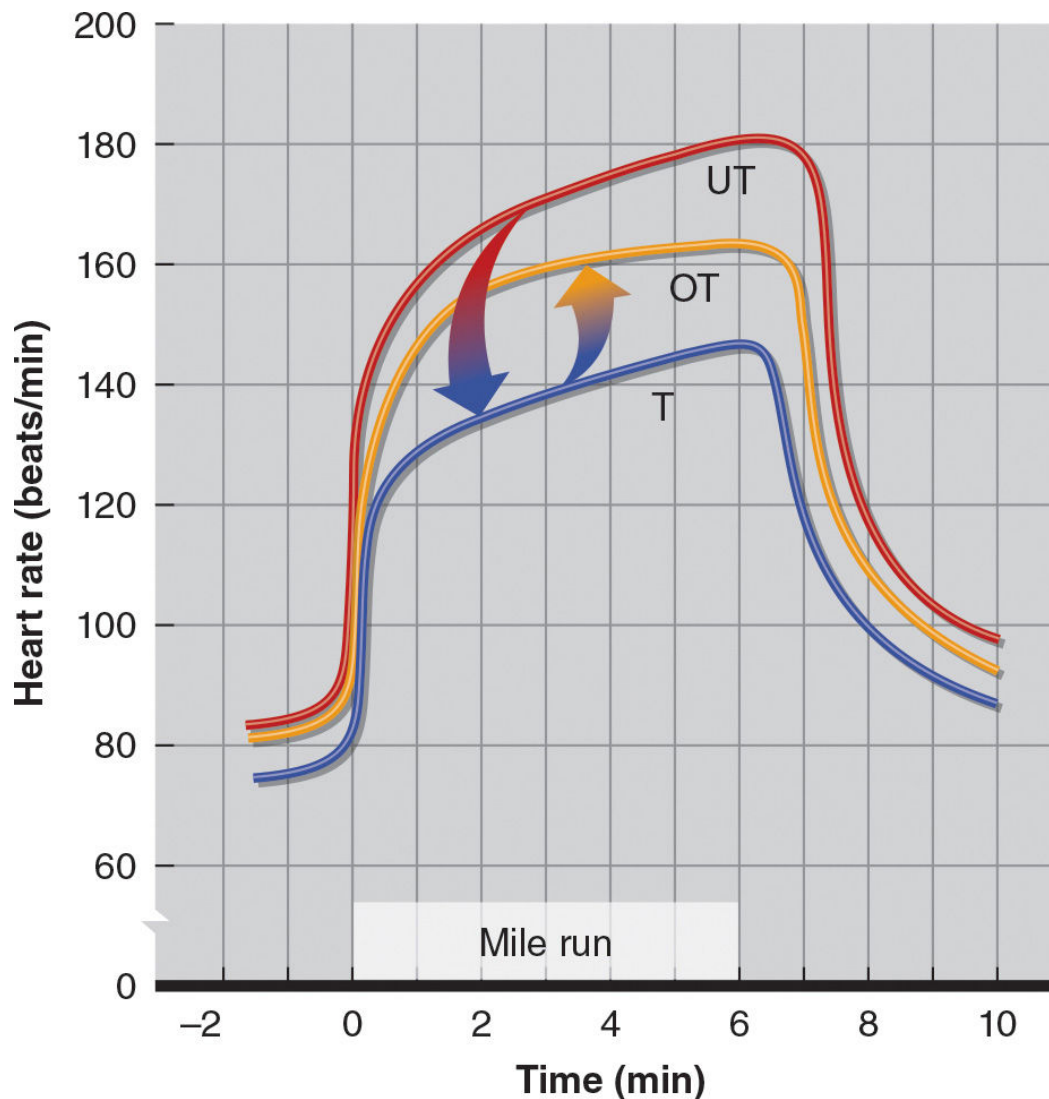


FIGURE 14.8 A runner's heart rate responses during a fixed-pace treadmill run at 10 mph (16 km/h) performed before training (UT), after training (T), and when the runner showed symptoms of overtraining (OT).

The distinction between healthy levels of exercise and exercise addiction is difficult to make. People who are addicted to exercise will continue to exercise regardless of physical injury, personal inconvenience, disruption to other areas of their lives (e.g., marriage or career), and lack of time for other important activities.¹⁵ Committed exercisers find a variety of rewards in their physical activity and view their exercise routines as an important but not all-defining part of their lives. Addicted exercisers identify exercise as a central part of their lives and exercise to avoid unpleasant emotions that would arise if they missed their exercise (e.g., guilt, anxiety). Addicted exercisers

may even consider exercise *the* most important part of their lives and the only way to elevate their mood or feel better. A 2013 review on the topic of exercise addiction¹⁵ notes that “exercisers must acquire a sense of life-balance while embracing an attitude conducive to sustainable long-term physical, psychological and social health outcomes”.

Prevention and Recovery

Recovery from the overtraining syndrome is possible with a marked reduction in training intensity or complete rest. Although most coaches recommend a few days of easy training, overtrained athletes require considerably more time for full recovery. This might necessitate the total cessation of training for a period of weeks or months. In some cases, counseling might be needed to help athletes cope with other stress in their lives that might contribute to this condition.

The best way to minimize the risk of overtraining is to follow periodization training procedures, alternating easy, moderate, and hard periods of training. Although individual tolerance varies tremendously, even the strongest athletes have periods when they are susceptible to the overtraining syndrome. As a rule, 1 or 2 days of intense training should be followed by an equal number of easy training days. Likewise, a week or two of hard training should be followed by a week of reduced effort with little or no emphasis on anaerobic exercise.

Endurance athletes (such as swimmers, cyclists, and runners) should pay particular attention to their caloric and carbohydrate intakes. Repeated days of hard training gradually reduce muscle glycogen. Unless these athletes consume extra carbohydrate during these periods, muscle and liver glycogen reserves can be depleted. As a consequence, the most heavily recruited muscle fibers are not able to generate the energy needed for exercise.

In Review

- Addicted exercisers identify exercise as a central part of their lives and exercise to avoid unpleasant emotions that would arise if they missed their exercise (e.g., guilt, anxiety).

- Athletes often overtrain in the erroneous belief that more training always produces further improvement. As their performance declines with overtraining, they train even harder in an effort to compensate. The importance of designing training programs to include rest and variations in training intensity and volume to avoid overtraining syndrome cannot be overstated.
 - Overtraining stresses the body beyond its capacity to adapt, decreasing physiological capacity and performance.
 - The symptoms of overtraining syndrome are subjective and vary from individual to individual; many also accompany regular training, making prevention or diagnosis of the overtraining syndrome difficult.
 - Possible explanations for the overtraining syndrome include changes in function of the autonomic nervous system, altered endocrine responses, suppressed immune function, and altered brain neurotransmitters.
 - Many potential signs and symptoms of overtraining have been proposed to diagnose overtraining in its earliest stages. However, at this time, the heart rate response to a fixed-pace exercise bout appears to be the easiest and most accurate technique. Performance decrements are also good indicators but often appear later in the syndrome.
 - Overtraining syndrome is treated by a marked reduction in training intensity or complete rest, for weeks or months. Prevention can best be accomplished through use of periodization training procedures that vary training intensity and volume.
 - For endurance athletes, it is important to ensure adequate calorie and carbohydrate intake to meet energy needs.
-

Tapering for Peak Performance

Peak performance requires maximal physical and psychological tolerance for the stress of the activity. But periods of intense training can reduce muscular strength, decreasing athletes' performance capacity. For this reason, to compete at their peak, many athletes reduce their training intensity and volume before a major competition to give their bodies a break from the rigors of intense training, a practice commonly referred to as **tapering**. The **taper period**, during which intensity and volume are reduced, provides adequate time for healing of tissue damage caused by intense training and for the body's energy reserves to be fully replenished. Tapering periods

range from 4 to 28 days or longer, depending on the sport, the event, and the athlete's needs. Tapering is not appropriate for all sports, particularly those in which competition occurs once a week or more frequently. Still, rested athletes generally perform better.

The most notable change during the taper period is a marked increase in muscular strength, which explains at least part of the performance improvement that occurs. It is difficult to determine whether strength improvements result from changes in the muscles' contractile mechanisms or improved muscle fiber recruitment. However, examination of individual muscle fibers taken from swimmers' arms before and after 10 days of intensified training showed that the type II (fast-twitch) fibers exhibited a significant reduction in their maximal shortening velocity.⁸ This change has been attributed to changes in the fibers' myosin molecules. In these cases, the myosin in the type II fibers became more like that in the type I fibers. We assume from these data that such changes in the muscle fibers cause the power loss that swimmers and runners experience during prolonged periods of intense training. We can also assume that the recovery of strength and power that occurs with tapering is linked to modifications of the muscles' contractile mechanisms. Tapering also allows time for the muscle to repair any damage incurred during intense training and for the energy reserves (i.e., muscle and liver glycogen) to be restored.

Although tapering is widely practiced in a variety of sports, many coaches fear that reduced training for such a long period before a major competition will decrease conditioning and impair performance. But numerous studies clearly show that this fear is unwarranted. Developing optimal $\dot{V}O_{2\max}$ initially requires a considerable amount of training, but once it has been developed, much less training is needed to maintain it at its highest level. In fact, the training level of $\dot{V}O_{2\max}$ can be maintained even when training frequency is reduced by two-thirds.¹¹

Runners and swimmers who reduce their training by about 60% for 15 to 21 days show no losses in $\dot{V}O_{2\max}$ or endurance performance.^{5,12} One study showed that swimmers' blood lactate concentrations after a standard swim were lower after a taper period than before. More important, the swimmers experienced a 3%

improvement in performance as a result of the reduced training and demonstrated an 18% to 25% increase in arm strength and power.⁵ In a study of distance runners, those runners who went through a 7-day taper decreased their running time in a 5 km time trial by 3% compared to no improvement in those who did not taper. Submaximal oxygen uptake during running at 80% $\dot{V}O_{2\max}$ was decreased by 6% in those who tapered, indicating a greater economy of effort. Blood lactate concentrations at 80% of $\dot{V}O_{2\max}$ were unchanged, as were $\dot{V}O_{2\max}$ and leg extension peak force.¹³

RESEARCH PERSPECTIVE 14.2

Peak Performance During the Taper Phase

Coaches of elite athletes have a common goal of designing a well-controlled training program to ensure that the maximal performance is achieved at major competitions. The best competition performances often occur following a taper phase, which is a progressive, nonlinear reduction of the training load in the period before competition. The main purpose of the taper is to reduce physiological and psychological stressors of previous training and remove residual fatigue so that sport performance can be optimized. However, there is relatively little scientific evidence available to guide coaches in prescribing appropriate tapering strategies for individual athletes. Most studies have used mathematical modeling simulations to predict the most ideal tapering strategies. Despite this limited support, it is common for athletes to undertake a period of overload training before tapering in an effort to maximize performance gains.

A 2014 study² described the relations between performance and training before and during a simulated taper. The investigators examined whether well-trained triathletes would demonstrate greater performance improvements than a control group during a simulated 4-week taper after completing 3 weeks of overload training. The investigators tested the hypothesis that completing overload training before the taper would allow larger performance gains, particularly in overreached athletes (those demonstrating decreased performance), but this would require a longer taper for performance compensation. The training of each triathlete was monitored for 11 weeks, which was divided into four distinct phases. The first two phases were identical for both the overload training and control groups: phase I was 3 weeks during which the subjects completed their own usual training and phase II was 1 week of moderate training load during which the subjects reduced habitual training volume by 30% while maintaining training intensity. During phase III, the control group repeated the training program as in phase I; however, the overload training group completed 3 weeks of training designed to deliberately

overreach the subjects (duration of each training session was increased by 30%). During phase IV, all participants completed a 4-week taper, where their normal training load was decreased by 40% each week. At the end of phases II and III, and each week during phase IV, subjects reported to the laboratory for a maximal oxygen consumption test.

In the overload training group of athletes, 10 triathletes developed clear symptoms of being functionally overreached (transient reduced performance associated with high perceived fatigue followed by the occurrence of a performance supercompensation), whereas 12 triathletes did not. The investigators found that greater gains in performance and maximal oxygen consumption were achieved when a higher training load was prescribed before the taper but not if functional overreaching occurs. Functional overreaching was associated with poor performance supercompensation during tapering and presented a greater risk for training maladaptation, including increased risk of injury or infection.



Practically, these results highlight the importance of carefully monitoring athletes during training, as well as during the taper. The investigators suggest that future studies should examine the effectiveness of different training models using early markers of functional overreaching to inform adjustments in training loads in an attempt to maximize training, tapering, and performance responses.

The effectiveness of various tapering strategies on improving sport performance is difficult to assess from any single study. One way to analyze data across many studies is to do a meta-analysis, which focuses on combining the results from a number of research studies in the hope of identifying patterns and contrasts among the various

results; the goal is to identify interesting relationships and conclusions from multiple studies. The effects of different types of tapering on competitive performance were assessed by meta-analysis of data from 27 studies.³ The authors analyzed tapering by determining the decrease in training intensity, volume, and frequency, as well as the pattern of tapering and duration of the tapering period. Most of the studies involved competitive swimming, running, and cycling. The optimal tapering strategy that emerged from this analysis was a tapering duration of 2 weeks over which the training volume is decreased by 41% to 60% without any change in intensity or frequency.

Unfortunately, little information is available to demonstrate the influence of tapering on performance in team sports and in long-duration endurance events such as cycling and marathon running. Before guidelines can be offered for athletes in these sports, research is needed to demonstrate that similar benefits can be generated by such periods of reduced training.

VIDEO 14.1 Presents Scott Trappe on tapering for peak sport performance.



In Review

- Many athletes decrease their training intensity and volume before a competition to increase strength, power, and performance capacity. This practice is called tapering.
- In some sports, tapering for competition is crucial to optimal performance. Reduced training volume and intensity, coupled with quality rest, are needed to

allow muscle to repair itself and to restore its energy reserves for competition.

- The optimal duration of the taper is between 4 and 28 days, or longer, and is dependent on the sport or event and the athlete's needs.
 - Muscular strength increases significantly during the tapering period.
 - Tapering allows time for the muscle to repair any damage incurred during intense training and for the energy reserves (i.e., muscle and liver glycogen) to be restored.
 - Less training is needed to maintain previous gains than was originally needed to attain them, so tapering does not decrease conditioning.
 - Aerobic performance may improve by an average of about 3% with proper tapering.
-

Detraining

Tapering, by reducing the training stimulus, can facilitate performance. What happens to highly conditioned athletes who have fine-tuned their performance abilities to a peak level but then come to the end of their competitive season? Many athletes welcome the opportunity to completely relax, purposely avoiding any strenuous physical activity. But how does physical inactivity affect highly trained athletes from a physiological standpoint?

Detraining is defined as the partial or complete loss of training-induced adaptations in response to either the cessation of training or a substantial decrement in the training load—in contrast to tapering, which is a gradual reduction of the peak training load over only a few days to a few weeks. Some of our knowledge about detraining comes from clinical research with patients who have been forced into inactivity because of injury or surgery. Most athletes fear that all they have gained through hard training will be lost during even a brief period of inactivity. But recent studies reveal that a few days of rest or reduced training will not impair and might even enhance performance, similar to what we see with tapering. Yet at some point, training reduction or complete inactivity will decrease physiological function and performance.

Muscular Strength and Power

When a broken limb is immobilized in a rigid cast, changes begin almost immediately in both the bone and the muscles. Within only a

few days, the cast that was applied tightly around the injured limb is loose. After several weeks, a large space separates the cast and the limb. Skeletal muscles undergo a substantial decrease in size, a process known as atrophy, when they remain inactive. Not surprisingly, muscle atrophy is accompanied by considerable loss of muscular strength and power. Total inactivity leads to rapid losses, but even prolonged periods of reduced activity lead to gradual losses that eventually can become quite significant.

Similarly, research confirms that muscular strength and power both are reduced once athletes stop training. The rate and magnitude of loss vary with the level of training. Highly skilled, accomplished weightlifters appear to have a rather rapid decline in strength within a few weeks of discontinuing intense training.⁹ With previously untrained people, strength gains can be maintained from several weeks up to over 7 months. In a study of young (20-30 years) and older (65-75 years) men and women who trained for 9 weeks, the increase in strength (1-repetition maximum) averaged 34% for the younger subjects and 28% for the older subjects, with no differences between men and women. After 12 weeks of detraining, none of the four groups had significant losses in strength from the end of the 9-week training program values. After 31 weeks of detraining, there was only an 8% loss in the younger subjects and a 13% decrease in the older subjects.¹⁶

A study with collegiate swimmers revealed that even with up to 4 weeks of inactivity, terminating training did not affect arm or shoulder strength.⁴ No strength changes were seen in these swimmers, whether they spent 4 weeks at complete rest or whether they reduced their training frequency to one or three sessions per week. But swimming power was reduced by 8% to about 14% during the 4 weeks of reduced activity, whether the swimmers underwent complete rest or merely reduced training frequency. Although muscular strength might not have diminished during the 4 weeks of rest or reduced training, the swimmers might have lost their ability to apply force during swimming, possibly attributable to a loss of skill.

The physiological mechanisms responsible for the loss of muscular strength as a consequence of either immobilization or inactivity are not clearly understood. Muscle atrophy causes a noticeable decrease

in muscle mass and water content, which could partly account for a loss in the development of maximal muscle fiber tension. Changes occur in the rates of protein synthesis and degradation as well as in specific fiber type characteristics. When muscles aren't used, the frequency of their neurological stimulation is reduced, and normal fiber recruitment is disrupted. Thus, part of the strength loss associated with detraining could result from an inability to activate some muscle fibers.

The retention of muscular strength, power, and size is extremely important for the injured athlete. The athlete can save much time and effort during rehabilitation by performing even a low level of exercise with the injured limb, starting in the first few days of recovery. Simple isometric contractions are very effective for rehabilitation because their intensity can be graded and they do not require joint movement. Any program of rehabilitation, however, must be designed in cooperation with the supervising physician and physical therapist.

Muscular Endurance

Muscular endurance decreases after only 2 weeks of inactivity. At this time, not enough evidence is available to establish whether this performance decrement results from changes in the muscle or from changes in cardiovascular capacity. In this section, we examine muscle changes that are known to accompany detraining and that could decrease muscular endurance.

The localized muscle adaptations that occur during periods of inactivity are well documented, but the exact role that these changes play in the loss of muscular endurance is unclear. We know from postsurgery cases that after a week or two of cast immobilization, the activities of oxidative enzymes such as succinate dehydrogenase (SDH) and cytochrome oxidase decrease by 40% to 60%. Data collected from swimmers, shown in [figure 14.9](#), indicate that the muscles' oxidative potential decreases much more rapidly than the subjects' maximal oxygen uptake with detraining. Reduced oxidative enzyme activity would be expected to impair muscular endurance, and this most likely relates to submaximal endurance capacity rather than to maximal aerobic capacity, or $\dot{V}O_{2\max}$.

In contrast, when athletes stop training, the muscles' glycolytic enzymes, such as phosphorylase and phosphofructokinase, change

little, if at all, for at least 4 weeks. In fact, Coyle and colleagues⁷ observed no change in glycolytic enzyme activities with up to 84 days of detraining compared with a nearly 60% decrease in the activities of various oxidative enzymes. This might at least partly explain why performance times in sprint events are unaffected by a month or more of inactivity, but the ability to perform longer endurance events may decrease significantly with as little as 2 weeks of detraining.

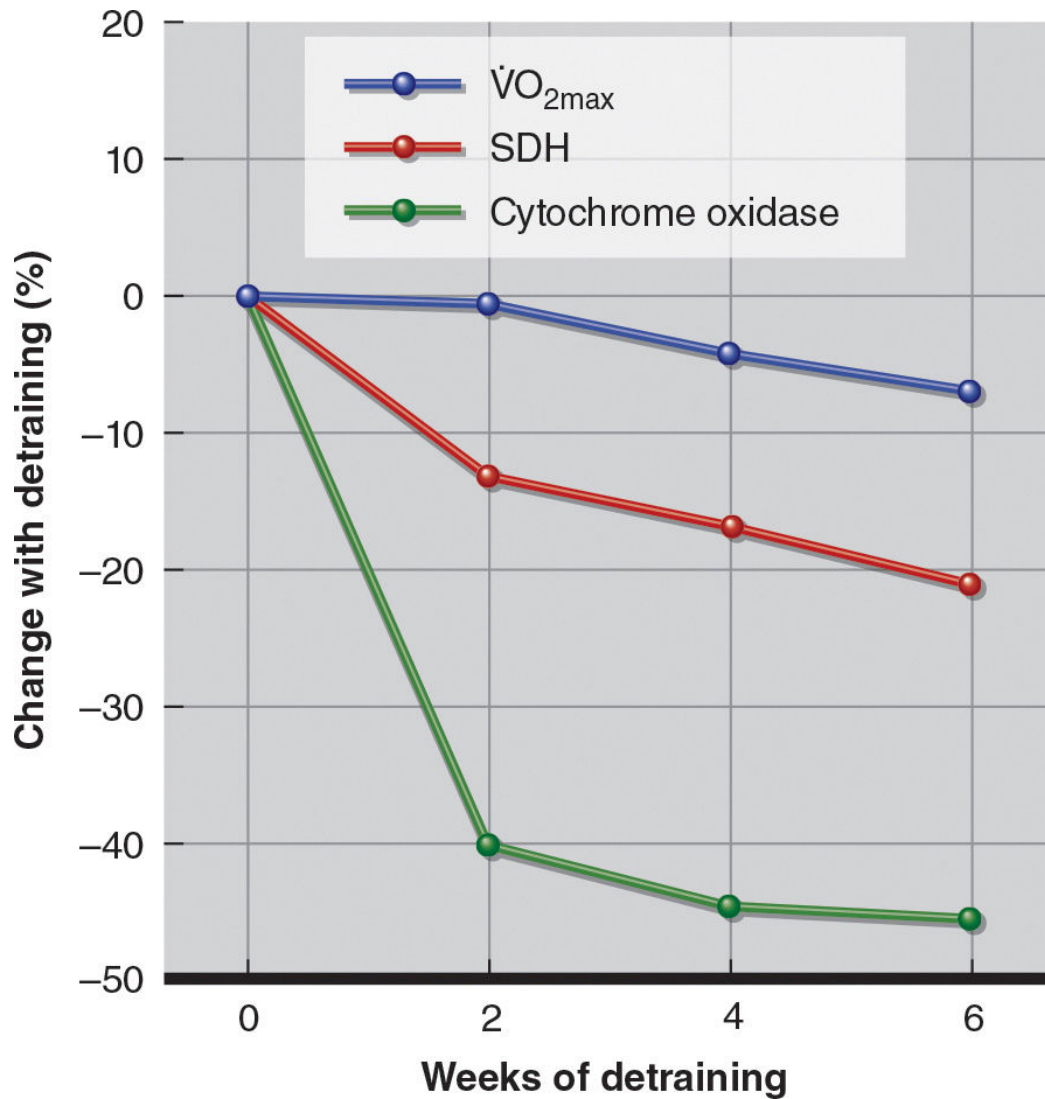


FIGURE 14.9 Percentage decreases in $\dot{V}O_{2\max}$, muscle succinate dehydrogenase (SDH) activity, and cytochrome oxidase activity during 6 weeks of detraining. These findings suggest that muscles experience a decline in metabolic potential, although tests of $\dot{V}O_{2\max}$ show little change over this period of detraining.

One notable change in muscle during detraining is a change in its glycogen content. Endurance-trained muscle tends to increase glycogen storage. But 4 weeks of detraining has been shown to decrease muscle glycogen by as much as 40%. [Figure 14.10](#) illustrates the decrease in muscle glycogen accompanying 4 weeks of detraining in competitive collegiate swimmers and in untrained subjects (serving as a time control). The untrained people showed no change in muscle glycogen content after 4 weeks of inactivity, but the swimmers' values, which were high initially, decreased until they were about equal to those of the untrained people. This indicates that the trained swimmers' improved capacity for muscle glycogen storage was reversed during detraining.

Measurements of blood lactate and pH after a standard work bout have been used to assess the physiological changes that accompany training and detraining. For example, a group of collegiate swimmers were required to perform a standard-paced, 200 yd (183 m) swim at 90% of their seasonal best following 5 months of training and then to repeat this test at the same absolute pace once a week for the following 4 weeks of detraining. The results are shown in [table 14.2](#). Blood lactate concentrations, measured immediately after this standard swim, increased from week to week during a month of inactivity. At the end of the fourth week of detraining, the swimmers' acid–base balance was significantly disturbed. This was reflected by a significant increase in blood lactate concentrations and a significant decrease in the concentrations of bicarbonate (a buffer).

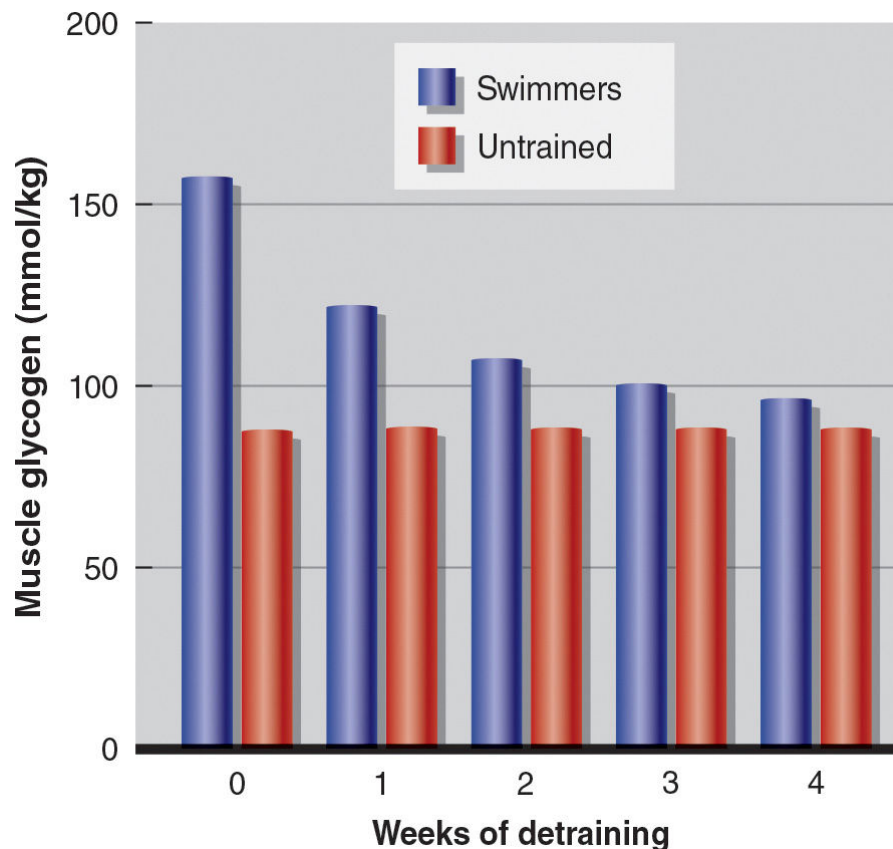


FIGURE 14.10 Changes in glycogen content of the deltoid muscle in competitive swimmers during 4 weeks of detraining. Muscle glycogen returned almost to the untrained concentration at the end of this period.

Speed, Agility, and Flexibility

Training produces less improvement in speed and agility than it does in strength, power, muscular endurance, flexibility, and cardiorespiratory endurance. Consequently, losses of speed and agility that occur with inactivity are relatively small, and peak speed and agility can be maintained with a limited amount of training. But this does not imply that the track sprinter can get by with training only a few days a week. Success in competition relies on factors other than basic speed and agility, such as correct form, skill, and the ability to generate a strong finishing sprint. Many hours of practice are required to tune performance to its optimal level, but most of this time is spent developing performance qualities other than speed and agility.

Flexibility, on the other hand, is lost rather quickly during inactivity. Stretching exercises should be incorporated into both in-season and

off-season training programs. It has been proposed that reduced flexibility increases athletes' susceptibility to serious injury.

TABLE 14.2 Blood Lactate, pH, and Bicarbonate (HCO_3^-) Values in Collegiate Swimmers Undergoing Detraining

Measurement	Weeks of detraining			
	0	1	2	4
Lactate (mmol/L)	4.2	6.3	6.8	9.7*
pH	7.26	7.24	7.24	7.18*
HCO_3^- (mmol/L)	21.1	19.5*	16.1*	16.3*
Swim time (s)	130.6	130.1	130.5	130.0

Note. Measurements were taken immediately after a fixed-pace swim. The values at week 0 represent the measurements taken at the end of 5 months of training. The values for weeks 1, 2, and 4 are the results obtained after 1, 2, and 4 weeks of detraining, respectively.

*Significant difference from the value at the end of training.

Cardiorespiratory Endurance

The heart, like other muscles in the body, is strengthened by endurance training. Inactivity, on the other hand, can substantially decondition the heart and the cardiovascular system. The most dramatic example of this is seen in a classic study conducted on subjects undergoing long periods of total bed rest; they weren't allowed to leave their beds, and physical activity was kept to an absolute minimum.¹⁹ Cardiovascular and metabolic function were assessed at a constant submaximal rate of work and at maximal rates of work both before and after the 20-day period of bed rest. The cardiovascular effects that accompanied bed rest included

- a considerable increase in submaximal heart rate,
- a 25% decrease in submaximal stroke volume,
- a 25% reduction in maximal cardiac output, and
- a 27% decrease in maximal oxygen consumption.

The reductions in cardiac output and $\dot{V}\text{O}_{2\text{max}}$ appear to result from reduced stroke volume; this seems to be largely attributable to a decreased plasma volume, with reductions in heart volume and ventricular contractility playing a smaller role.

It is interesting that the two most highly conditioned subjects in this study (the two who had the highest $\dot{V}\text{O}_{2\text{max}}$ values) experienced greater decrements in $\dot{V}\text{O}_{2\text{max}}$ than the three less fit people, as shown in [figure 14.11](#). Furthermore, the untrained subjects regained their

initial conditioning levels (before bed rest) in the first 10 days of reconditioning, but the well-trained subjects needed about 40 days for full recovery. This suggests that highly trained individuals cannot afford long periods with little or no endurance training. The athlete who totally abstains from endurance training at the completion of the season will experience great difficulty getting back into top aerobic fitness when the new season begins.

Inactivity can significantly reduce $\dot{V}O_{2\max}$. How much activity is needed to prevent such considerable losses of physical conditioning? Although a decrease in training frequency and duration reduces aerobic capacity, the losses are significant only when frequency and duration are reduced by two-thirds of the regular training load.

However, training intensity apparently plays a more crucial role in maintaining aerobic power during periods of reduced training. Training at 70% $\dot{V}O_{2\max}$ appears to be necessary to maintain maximal aerobic capacity.¹¹

Detraining During Spaceflight

As astronauts orbit the earth, they are in an environment where the gravitational forces are considerably less than those on Earth, that is, microgravity. While astronauts experience a sense of weightlessness in orbit, the gravitational forces do not reach 0 g. During an extended stay in microgravity, astronauts undergo physiological changes that are nearly identical to those of detraining. But what could be perceived as maladaptations on Earth might in fact be necessary adaptations to accommodate the microgravity of space. Let's briefly review the changes that take place when astronauts leave the 1 g environment on Earth to spend weeks or months in space.

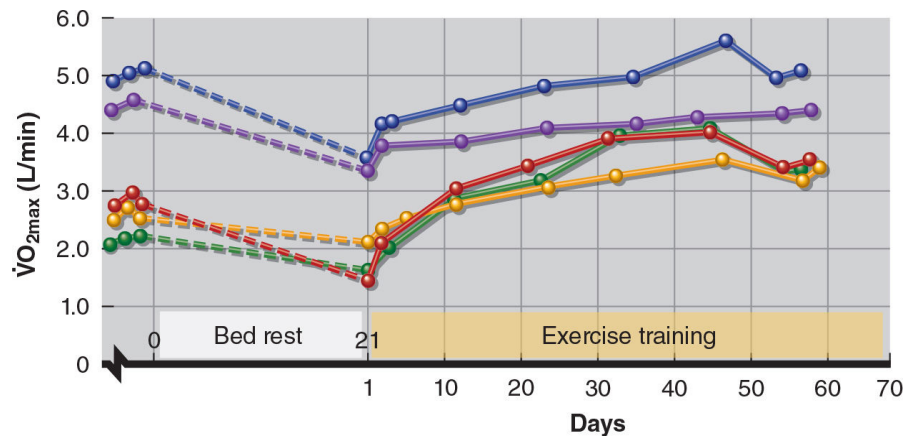


FIGURE 14.11 Changes in $\dot{V}O_{2\max}$ with 20 days of bed rest for five individual subjects. Note that the subjects who were least fit (lowest $\dot{V}O_{2\max}$ values) at the start of bed rest showed smaller decrements with inactivity and greater gains when they trained after bed rest. Highly fit individuals, on the other hand, were far more affected by the period of inactivity.

Adapted by permission from B. Saltin et al., "A Longitudinal Study of Adaptive Changes in Oxygen Transport and Body Composition," *Circulation* 38, no. VII (1968): 1-78.

Muscle mass and strength decline in microgravity, particularly in the postural muscles, that is, those muscles that maintain the body upright countering the force of gravity. The cross-sectional area of both type I and type II muscle fibers is decreased as well. The extent of decline depends on the muscle group, the duration of the flight, and the type and extent of the in-flight exercise program. Microgravity also affects bone, with average bone mineral losses of about 4% in the weight-bearing bones; the magnitude of this loss depends on the length of exposure to microgravity.

The cardiovascular system also undergoes major adaptations to microgravity. When the body is in microgravity, there is a reduction in hydrostatic pressure so that blood no longer pools in the lower extremities as it does at 1 g. As a consequence, more blood returns to the heart, which leads to transient increases in stroke volume. Reductions in plasma volume occur over time but are likely due to reduced fluid intake rather than increased production of urine (diuresis) by the kidneys. Transcapillary fluid shifts between the microcirculation and surrounding tissues can also account for some of the reduction in plasma volume, most likely upper body capillary filtration; for example, blood relocates into the facial tissues, creating a puffy facial appearance. Red cell mass also decreases, so total blood volume is reduced as well. The reduced blood volume serves

astronauts well while they remain in microgravity. But it presents a serious problem on their return to a 1 g environment, where the body is again subjected to the hydrostatic pressure effect. Astronauts have experienced postural (orthostatic) hypotension and fainting during their first few hours back in a normal 1 g environment because their blood volume was insufficient to meet all their circulatory needs.

RESEARCH PERSPECTIVE 14.3

Disturbed Sleep and Increased Illness in Overreached Athletes

In an attempt to improve the physiological adaptations to training and maximize sport performance, most athletes increase exercise intensity and volume at some point during training. However, this practice may also lead to functional overreaching and compromises short-term performance outcomes, especially when the recovery time is insufficient. Although the causes of functional overreaching remain incompletely understood, it is well known that adequate passive rest and sufficient sleep promote enhanced recovery and reduced fatigue. The first study to look at the link between disruptions in sleep and functional overreaching was conducted in 2014 to determine whether changes in objective sleep parameters were evident between a healthy group of athletes and a group of athletes developing functional overreaching following overload training.¹⁰

During the study, performance was assessed by measurement of $\dot{V}O_{2\max}$ and sleep was monitored using actigraphy. As expected, there was a progressive decline in indices of sleep quality and small reductions in sleep quantity during the overload training period in athletes who developed functional overreaching. In these athletes, alterations in sleep metrics were reversed during the subsequent taper phase of training. In the control group, no athletes developed any symptoms of training intolerance. However, it remains unclear whether poor sleep was a consequence or a cause of functional overreaching; for these determination, additional studies must be conducted.

Maximal aerobic power ($\dot{V}O_{2\max}$) is generally reduced immediately postflight, likely due to the reduction in plasma volume and leg strength during flight. However, data are limited on directly measured $\dot{V}O_{2\max}$ in astronauts preflight, during flight, and postflight. Head-down tilt (-6°) bed rest, used as an Earth-based model of spaceflight, shows consistent reductions in $\dot{V}O_{2\max}$ that are associated with

reductions in total blood volume, plasma volume, and, consequently, maximal stroke volume. The head-down tilt model has been shown to provide $\dot{V}O_{2\max}$ (or $\dot{V}O_{2\text{peak}}$) data comparable to actual pre- to postflight changes.²²

Importantly, having an understanding of the general decline in physiological function that occurs during spaceflight has led the scientific and medical community to the realization that in-flight exercise programs are essential to preserve the long-term health of astronauts. Research is now under way to design the most appropriate programs and exercise equipment to meet this objective.

In Review

- The body rapidly loses many of the benefits of training if training is discontinued. Some minimal level of training is necessary to prevent these losses. Research indicates that at least three training sessions per week at an intensity of at least 70% $\dot{V}O_{2\max}$ are needed to maintain aerobic conditioning.
- Detraining is defined as the partial or complete loss of training-induced adaptations in response to either the cessation of training or a substantial decrement in the training load.
- Detraining causes muscle atrophy, which is accompanied by losses of muscular strength and endurance. However, muscles require only minimal stimulation to retain these qualities during periods of reduced activity.
- Muscular endurance decreases after only 2 weeks of inactivity. Possible explanations for this are
 1. decreased oxidative enzyme activity,
 2. decreased muscle glycogen storage, or
 3. disturbance of acid–base balance.
- Detraining-induced losses in speed and agility are small, but flexibility seems to be lost quickly.
- With detraining, losses of cardiorespiratory endurance are much greater than losses of muscular strength, power, and endurance over the same time period.
- To maintain cardiorespiratory endurance, training must be conducted at least three times per week, and training intensity should be at least 70% of $\dot{V}O_{2\max}$.
- Prolonged exposure to microgravity in space causes physiological changes that are nearly identical to those of detraining. While these are generally

maladaptations on Earth, in space they serve as positive physiological adaptations that lead to a better ability to tolerate microgravity.

IN CLOSING

In this chapter, we have examined how the quantity of training can affect performance. We saw that too much training, in the form of either excessive training or overtraining, can actually impair performance. Then we looked at the effects of too little training—detraining—as a result of either inactivity or immobilization after an injury. Finally, we saw that with detraining, many of the gains achieved during regular training are quickly lost, especially cardiovascular endurance.

Now that we have dispelled the myth that more training always means better performance, it makes sense to ask about other ways in which athletes can try to optimize their performance. In the next chapter, we turn our attention to optimal body composition and nutrition for the serious athlete.

KEY TERMS

acute overload
detraining
excessive training
immune function
overreaching
overtraining
overtraining syndrome
periodization
tapering
taper period
undertraining

STUDY QUESTIONS

1. Describe the model used to optimize training. Define the terms *undertraining*, *acute overload*, *overreaching*, and *overtraining*.
2. What is excessive training? How does it relate to the model for optimizing training?
3. Distinguish between traditional and block forms of training periodization. What are the advantages and disadvantages of each?
4. Define and describe the overtraining syndrome. What are the general symptoms of the overtraining syndrome? How do these differ between sympathetic and parasympathetic overtraining?
5. What distinguishes a committed exerciser from one who may be addicted to exercise?

6. How might the hypothalamus be involved in the overtraining syndrome? What role might cytokines play?
7. Describe the relationship between physical activity and immune function and disease susceptibility.
8. What appears to be the best predictor of the overtraining syndrome?
9. How do we treat the overtraining syndrome?
10. What physiological changes occur during the taper period that can be credited with improvements in performance?
11. What alterations occur in strength, power, and muscular endurance with physical detraining?
12. What alterations occur in speed, agility, and flexibility with physical detraining?
13. What changes occur in cardiovascular function as one becomes deconditioned?
14. What similarities do we see between spaceflight and detraining?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

[**www.HumanKinetics.com/PhysiologyOfSportAndExercise**](http://www.HumanKinetics.com/PhysiologyOfSportAndExercise)

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.

